الكتابة الاحترافية بإستخدام نظام لاتك (لاتخ) Latex

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محاضرة حول
الكتابة الاحترافية باستخدام نظام
LaTeX

يـلـقـيـهـا

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المقدمة

$\LaTeX$ ويلفظ باللغه العربية (لاتخ) أو (لاتك) : هو نظام مبتكر يهدف لتنضيد النصوص العادية و نصوص الدساتير الرياضية، يحظى باستخدام واسع في أوساط الجاليات العلمية و الأكاديمية بما يوفره من سهولة تحرير و بساطة صيغة الأوامر والإيعازات، اللاتك ليس لغة برمجة بل لغة تنضيد و تنسيق مرنة.

بعض من التاريخ...

لاتك LaTeX ، تمّ تصميمه بادئ الأمر من قبل Leslie Lamport سنة 1985، وكان امتدادا مبسّطا لبرنامج تك TeX الذي صمّمه أستاذ رياضيّات ألماني، هو دونالد كنوث Donald E.Knuth عام 1978، لكتابة المقالات العلميّة وكتب الرّياضيّات بمعادلاتها ودساتيرها. يُعدّ لاتخ LaTeXواجهة التّخاطب بين المُستخدم وبرنامج تك TeX.

المقدمة

ظلّ استعمال هذا البرنامج ممكنا فقط في كتابة المقالات والكتب باللّغات التي تُكتب من اليسار إلى اليمين، (LTR) وقد استعمله، ويستعمله حالياً، مئات الآلاف من الباحثين والمهندسين والطّلاب في كتابة الملايين من كتبهم وتقاريرهم وأطروحاتهم، وقد زوّدوه بمكتبات توسّع من إمكانيّاته. لكنّ لاتك ظلّ أعجمياً لايفهم العربيّة، إلى أن تصدّى أحد المبرمجين لجعل لاتك يفهم العربيّة، وهو السيّد Klaus Lagally من جامعة شتوتغارت الألمانيّة، الذي زوّد لاتك بمكتبة دعاها ArabTeX، بيد أنّ هذه التّوسعة لم تشمل كلّ إمكانيّات لاتك، والتّعامل معها لايزال صعبا معقّدا.

أُصدرت حديثاً نسخة جديدة من لاتك تدعى كزيلاتك XeLaTeX تفهم النّصوص المرمّزة بنظام UTF8، ويمكنها أن تنسّق النّص من اليمين إلى اليسار وهي متأقلمة مع كلّ المكتبات المخصّصة لـ LaTeX. لكنّ التعامل مع هذه النسخة باللّغة العربيّة يوجب إضافة مكتبة خاصّة، تصدّى لهذه المهمة د. مصطفى العليوي من المعهد العالي للعلوم التطبيقية والتكنولوجيا، وأضاف مكتبة، سمّاها XeArabic، تبسّط استعمال XeLaTeX باللغة العربية وتختصره بشكل كبير، وذلك من أجل كتابة المقالات العلميّة والكتب والمحاضرات العربية بواسطة XeLaTeX .

كيفية تثبيت برنامج لاتك

1- تحميل وتثبيت البرنامج MikeTex 2.9 عن طريق هذا الرابط

http://miktex.org/2.9/setup وتختار نوع فايل التنصيب حسب نوع الوندوز الموجود في كومبيوترك أما 64 أو 32 بت.

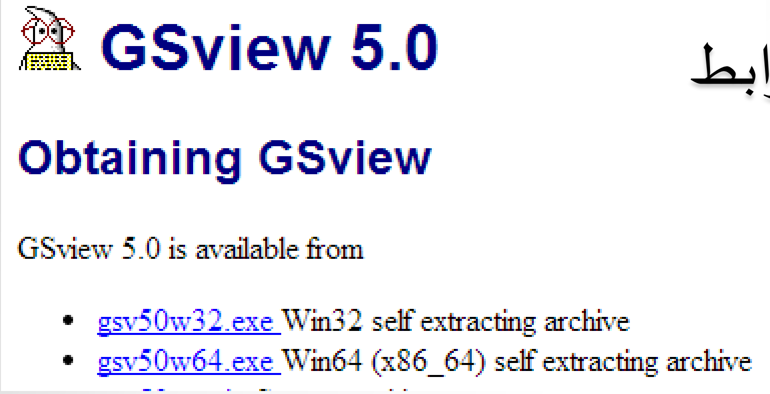
2- تحميل وتثبيت البرنامج Ghostscript 9.10 عن طريق هذا الرابط http://www.ghostscript.com/download/gsdnld.html

وطبعا تختار أما 64 أو 32 بت حسب نظام الوندوز المثبت في كومبيوترك.

License / Platform	AGPL GNU Affero General Public License	ARTIFEX Software Inc. Artifex Commercial License
Ghostscript 9.10 for Windows (32 bit)	Ghostscript GPL Release	Ghostscript Commercial License
Ghostscript 9.10 for Windows (64 bit)	Ghostscript GPL Release	Ghostscript Commercial License

3- تحميل وتثبيت البرنامج GSView 5.0 من هذا الرابط

. http://pages.cs.wisc.edu/~ghost/gsview/get50.htm



3- تحميل وتثبيت برنامج WinEdt وهذا البرنامج ليس مجانا لذلك قمت بتحميلة على موقع ميديا فاير ويمكن الحصول عليه من هذا الرابط

http://www.mediafire.com/download/iqaw4lol3e589hq/WinEdt.rar

بعد تحميل البرنامج تقوم بفك الضغط عن طريق برنامج winrar ومن ثم يتولد فولدر أسمه WinEdt بعد فك الضغط في داخل الفولدر تجد العديد من الفايلات تختار الفايل الذي أسمه winedt53 وتثبته ومن ثم تسجل البرنامج من خلال المعلومات التالية

name : Hard Wisdom

sn : 11353621062783O983O

لماذا نستخدم لاتك بدلا من محرر النصوص وورد ؟

- ❖ مخرجات طباعة المعادلات الرياضية ذات وضوح ودقة عالية وأبعاد متناسقة أفضل بكثير من وورد.

(لاتك)

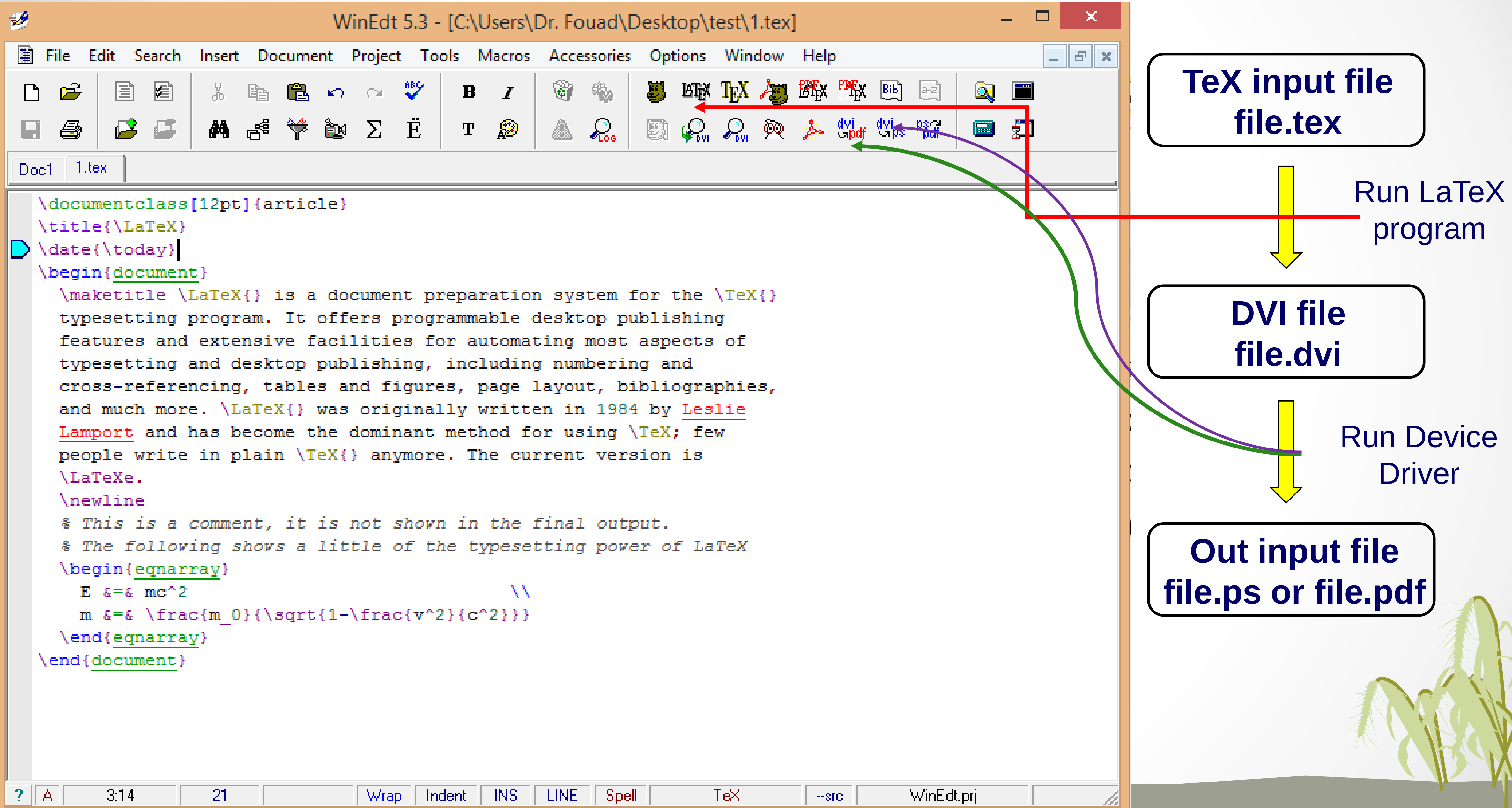
$$|F_J(q)|^2 = \frac{4\pi}{Z^2(2J_i+1)} \left| \sum_{T=0,1} \begin{pmatrix} T_f & T & T_i \\ -T_z & 0 & T_z \end{pmatrix} \right|^2$$
$$\times |\langle \alpha_2 ||| T_\Lambda ||| \alpha_1 \rangle|^2 \, |F_{c.m}(q)|^2 \, |F_{f.s}(q)|^2$$

(وورد)

$$|F_J(q)|^2 = \frac{4\pi}{Z^2(2J_i+1)} \left| \sum_{T=0,1} \begin{pmatrix} T_f & T & T_i \\ -T_z & 0 & T_z \end{pmatrix} \right|^2$$
$$\times \left| \langle \alpha_2 \| T_\Lambda \| \alpha_1 \rangle \right|^2 |F_{c.m}(q)|^2 |F_{f.s}(q)|^2$$

- ❖ سهل الأستخدام في كتابة المعادلات الرياضية ومهما كانت صعوبتها وكثرة الرموز والمتغيرات الموجود فيها.
- ❖ لا يعتمد على أي نظام تشغيل (أي أنه يعمل على كل الأنظمة المتوفرة) (وندوز، لينوكس ، ماك ... الخ).
- ❖ الأستقرارية العالية وأي فايل مكتوب بنظام لاتك يعمل وعلى أي أصدار بعكس الوورد فالفايلات التي كتبت بنظام 2013 لا تفتح بنظام 2003 وما دون ما لم تخزن بتلائم مع الأصدار 2003.
- ❖ معظم المجلات العلمية العالمية الرصينه في تصنيف ثوماس رويترز لها تنسيقات لاتك الخاصه بها والتي يمكن أن تحمل من موقع المجلات وتستعمل بشكل مباشر لأختصار الوقت والجهد.

شرح محرر نص لاتك WinEdt



البنية الأساسية لمستند (لاتك)

\documentclass [12pt]{article}

Define the types of the document (article, book, thesis …)

\usepackage {color}
\usepackage {graphicx}

Preamble. Incorporate packages or define macros here

\begin{document}

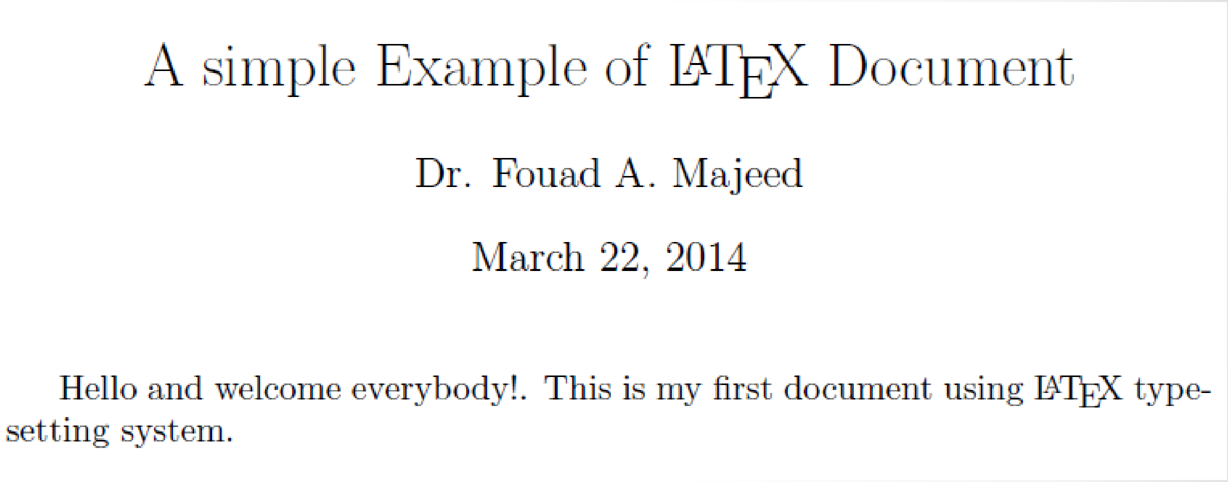
Main body, stuff to be printed, title, authors, abstract, sections, references, ….

\end{document}

مثال بسيط لمستند لاتك

```
\documentclass{article}
\title{A simple Example of \LaTeX\ Document}
\author{Dr. Fouad A. Majeed}
\date{\today}
\begin{document}
\maketitle

Hello and welcome everybody!. This is my first document using
\LaTeX\ typesetting system.
\end{document}
```



A simple Example of LaTeX Document

Dr. Fouad A. Majeed

March 22, 2014

Hello and welcome everybody!. This is my first document using LaTeX typesetting system.

أيعازات تنسيق النص

- Start with `\begin{document}`
- End with `\end{document}`
- Typesetting Text
  - `\\` or `\newline` and `\newpage`
  - Quotations
  - Bold `\textbf{...............}` or `\bf`
  - Italics `\emph{...........}` or `\textit{........}` or \it
  - Underline `\underline{...........}` or `\ul`

- Sections
  - `\section{Latex is Great}` = 1. Latex is Great
  - `\subsection{Why Latex is Great}` = 1.1 Why Latex is Great
  - `\subsubsection{Reason One}` = 1.1.1 Reason One
  - `\appendix` - changes numbering scheme
  - `\chapter{…}` - To be used with book and report document classes
- Titles, Authors and others
  - `\title{…}` `\author{…}`
  - `\footnote{…}`

- `\maketitle` - Display Title and Author
- `\tableofcontents` - generates TOC
- `\listoftables` - generates LOT
- `\listoffigures` - generates LOF
- Itemize
  - Use either *enumerate*, *itemize* or *description.*\
- Source
  - `\begin{itemize}`
  - `\item Apple`
  - `\item Orange`
  - `\end{itemize}`

- Apple
- Orange
- Mango

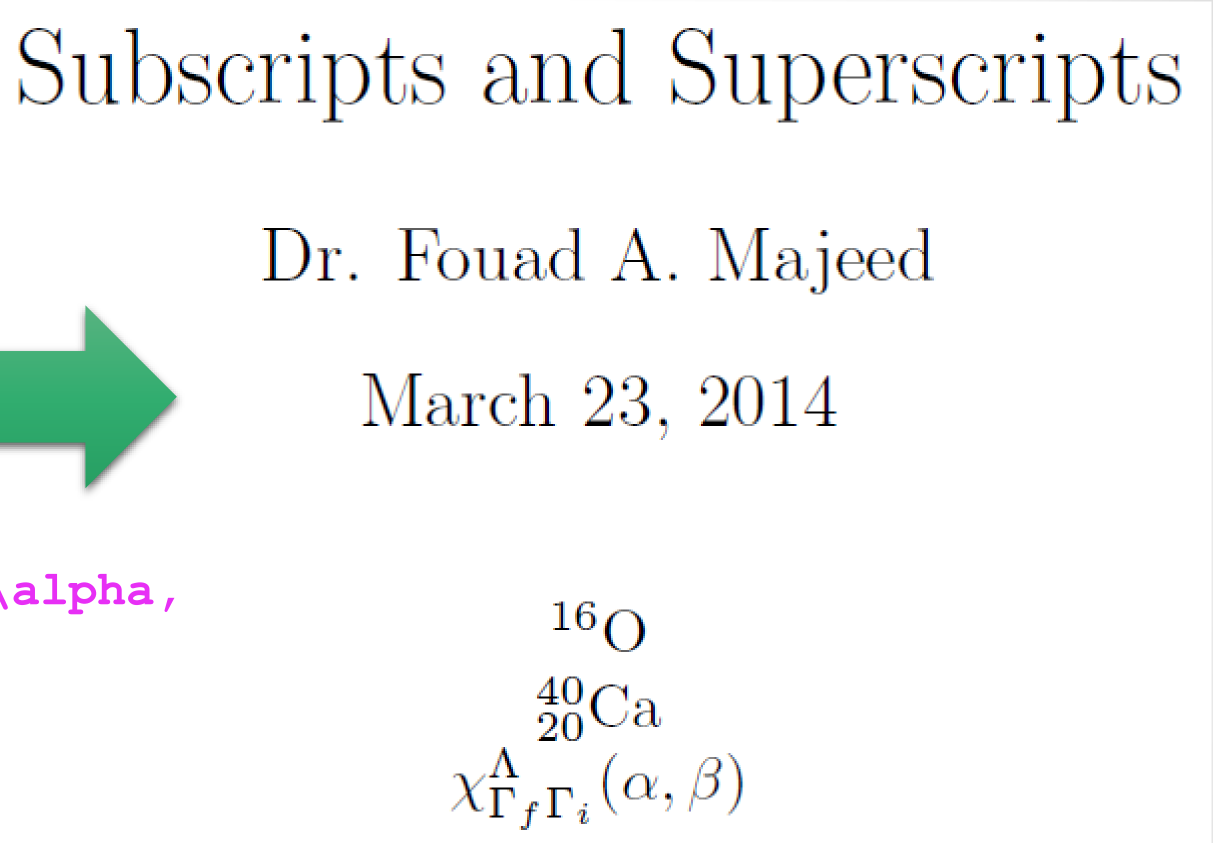
Subscripts and Superscpripts

```
\documentclass{article}
\title{Subscripts and Superscripts }
\author{Dr. Fouad A. Majeed}
\date{\today}
\begin{document}
\maketitle
\begin{center}
 $^{16}$O

 $^{40}_{20}$Ca

 $\chi^{\Lambda}_{\Gamma_{f}\Gamma_{i}}(\alpha,
\beta)$
\end{center}


\end{document}
```



Subscripts and Superscripts

Dr. Fouad A. Majeed

March 23, 2014

^{16}O

$^{40}_{20}$Ca

$\chi^{\Lambda}_{\Gamma_{f}\Gamma_{i}}(\alpha, \beta)$

أنشاء جدول بسيط في لاتك

```
\begin{tabular}{|l|r|c|}
\hline
Date & Price & Size \\
\hline
Yesterday & 5 & big \\
\hline
Today & 3 & small \\
hline
\end{tabular}
```

Date	Price	Size
Yesterday	5	big
Today	3	small

جدول أكثر تعقيدا

```
\begin{table}[ht]
\caption{Nonlinear Model Results} % title of Table
\centering                        % used for centering table
\begin{tabular}{c c c c}           % centered columns (4
columns)
\hline\hline %inserts double horizontal lines
Case & Method\#1 & Method\#2 & Method\#3 \\ [0.5ex] % inserts
table
%heading
\hline % inserts single horizontal line
1 & 50 & 837 & 970 \\ % inserting body of the table
2 & 47 & 877 & 230 \\
3 & 31 & 25 & 415 \\
4 & 35 & 144 & 2356 \\
5 & 45 & 300 & 556 \\ [1ex] % [1ex] adds vertical space
\hline %inserts single line
\end{tabular}
\label{table:nonlin} % is used to refer this table in the text
\end{table}
```

Table 1: Nonlinear Model Results

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415
4	35	144	2356
5	45	300	556

أدراج الأشكال والصور

```
\documentclass[12pt]{article}
\usepackage{graphicx}

\begin{document}

\listoffigures

\section{Introduction}

\begin{figure}[hb]
  \centering
  \includegraphics[width=0.5\textwidth]{leopard.jpg}
  \caption{The snow leopard (Panthera uncia or Uncia
uncia) is a moderately large cat native to the mountain
ranges of Central
  Asia.}
\end{figure}

\end{document}
```

List of Figures

1 Introduction

Figure 1: The snow leopard (Panthera uncia or Uncia uncia) is a moderately large cat native to the mountain ranges of Central Asia.

وبنفس الأسلوب يدرج الشكل العلمي

```
\documentclass[12pt]{article}
\usepackage{graphicx}

\begin{document}

\listoffigures

\section{Introduction}

\begin{figure}[hb]
  \centering
  \includegraphics[width=0.5\textwidth]{fig1.eps}
  \caption{The longitudinal $C0+C2$ form factor
for the isoscalar $3^{+}_{g.s.}$ (0.0 MeV)
transition in $^{10}$B compared with the
experimental data.}
\end{figure}

\end{document}
```

List of Figures

1 Introduction

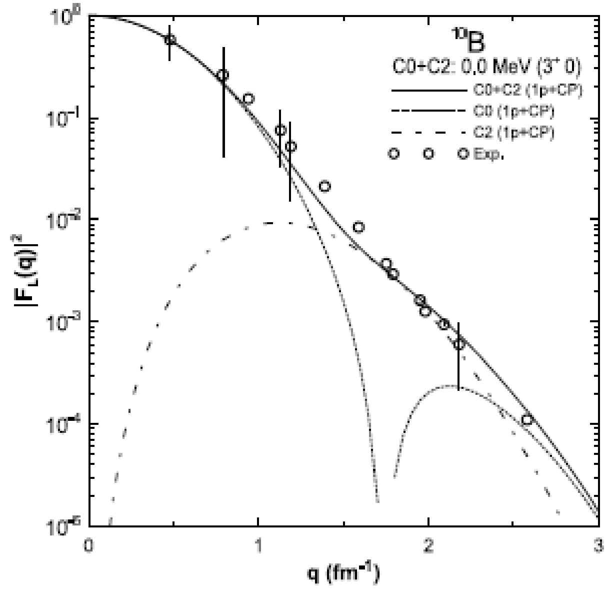


Figure 1: The longitudinal $C0+C2$ form factor for the isoscalar $3^+_{g.s.}$ (0.0 MeV) transition in ^{10}B compared with the experimental data.

كتابة المعادلات الرياضية بنظام لاتك

```latex
\documentclass[12pt]{article}
\usepackage{amsmath}
\title{Writing equations in \LaTeX }
\author{Dr. Fouad A. Majeed}
\begin{document}
\maketitle
\begin{align}
\cos^2\psi+\sin^2\psi & = 1, \\
\cosh^2\omega-\sinh^2\omega &=1.
\end{align}
\begin{equation}
F_{n} = \begin{cases}
0 & \text{if $n = 0$;}\\
1 & \text{if $n = 1$;}\\
F_{n - 1} + F_{n - 2} & \text{if $n > 0$.}
\end{cases}
\end{equation}
\[
\det \mathbf{K} (t = 1, t_{1}, \dots, t_{n}) = \sum_{I
\in
\mathbf{n} }(-1)^{|I|} \prod_{i \in I} t_{i} \prod_{j
\in I}
(D_{j} + \lambda_{j} t_{j}) \det \mathbf{A}^{(\lambda)}
(\overline{I} | \overline{I}) = 0
\]
```

Writing equations in LaTeX

Dr. Fouad A. Majeed

March 24, 2014

$$\cos^2\psi+\sin^2\psi = 1, \quad (1)$$

$$\cosh^2\omega-\sinh^2\omega =1. \quad (2)$$

$$F_{n} = \begin{cases} 0 & \text{if } n = 0;\\ 1 & \text{if } n = 1;\\ F_{n-1} + F_{n-2} & \text{if } n > 0. \end{cases} \quad (3)$$

$$\det \mathbf{K}(t = 1, t_{1}, \dots, t_{n}) = \sum_{I \in \mathbf{n}}(-1)^{|I|} \prod_{i \in I} t_{i} \prod_{j \in I} (D_{j} + \lambda_{j} t_{j}) \det \mathbf{A}^{(\lambda)}(\overline{I} | \overline{I}) = 0$$

تكملة كتابة المعادلات الرياضية

```
\mathbf{A} =
\begin{pmatrix}
\dfrac{\varphi \cdot X_{n, 1}} {\varphi_{1} \times
\varepsilon_{1}} & (x + \varepsilon_{2})^{2} & \cdots &
(x +
\varepsilon_{n - 1})^{n - 1}
& (x + \varepsilon_{n})^{n}\\
\dfrac{\varphi \cdot X_{n, 1}} {\varphi_{2} \times
\varepsilon_{1}} & \dfrac{\varphi \cdot X_{n, 2}}
{\varphi_{2}
\times \varepsilon_{2}} & \cdots & (x + \varepsilon_{n
- 1})^{n -
1}
& (x + \varepsilon_{n})^{n}\\
\hdotsfor{5}\\
\dfrac{\varphi \cdot X_{n, 1}} {\varphi_{n} \times
\varepsilon_{1}} & \dfrac{\varphi \cdot X_{n, 2}}
{\varphi_{n}
\times \varepsilon_{2}} & \cdots & \dfrac{\varphi \cdot
X_{n, n -
1}} {\varphi_{n} \times \varepsilon_{n - 1}} &
\dfrac{\varphi\cdot
X_{n, n}} {\varphi_{n} \times \varepsilon_{n}}
\end{pmatrix}
+ \mathbf{I}_{n}
\]
\end{document}
```

$$\mathbf{A} = \begin{pmatrix} \dfrac{\varphi \cdot X_{n,1}}{\varphi_1 \times \varepsilon_1} & (x+\varepsilon_2)^2 & \cdots & (x+\varepsilon_{n-1})^{n-1} & (x+\varepsilon_n)^n \\ \dfrac{\varphi \cdot X_{n,1}}{\varphi_2 \times \varepsilon_1} & \dfrac{\varphi \cdot X_{n,2}}{\varphi_2 \times \varepsilon_2} & \cdots & (x+\varepsilon_{n-1})^{n-1} & (x+\varepsilon_n)^n \\ \hdotsfor{5} \\ \dfrac{\varphi \cdot X_{n,1}}{\varphi_n \times \varepsilon_1} & \dfrac{\varphi \cdot X_{n,2}}{\varphi_n \times \varepsilon_2} & \cdots & \dfrac{\varphi \cdot X_{n,n-1}}{\varphi_n \times \varepsilon_{n-1}} & \dfrac{\varphi \cdot X_{n,n}}{\varphi_n \times \varepsilon_n} \end{pmatrix} + \mathbf{I}_n$$

طرق كتابة المصادر في نظام لاتك

الطريقة رقم (1)

\cite{Boney96}

```latex
\begin{document}
\maketitle

\section{Introduction}

An audio signal is a representation of sound, typically as an
electrical voltage. Audio signals have frequencies in the audio
frequency range of roughly 20 to 20,000 Hz (the limits of human
hearing)~\cite{Boney96}.

Audio signals may be synthesized directly, or may originate at a
transducer such as a microphone, musical instrument pickup,
phonograph cartridge, or tape head. Loudspeakers or headphones
convert an electrical audio signal into sound. Digital
representations of audio signals exist in a variety of
formats~\cite{MG,HK,Pan}.

\begin{thebibliography}{100} % 100 is a random guess of the total
number of
%references
\bibitem{Boney96} Boney, L., Tewfik, A.H., and Hamdy, K.N., ``Digital
Watermarks for Audio Signals," \emph{Proceedings of the Third IEEE
International Conference on Multimedia}, pp. 473-480, June 1996.
\bibitem{MG} Goossens, M., Mittelbach, F., Samarin, \emph{A LaTeX
Companion}, Addison-Wesley, Reading, MA, 1994.
\bibitem{HK} Kopka, H., Daly P.W., \emph{A Guide to LaTeX},
Addison-Wesley, Reading, MA, 1999.
\bibitem{Pan} Pan, D., ``A Tutorial on MPEG/Audio Compression,"
\emph{IEEE
Multimedia}, Vol.2, pp.60-74, Summer 1998.
\end{thebibliography}
\end{document}
```

1 Introduction

An audio signal is a representation of sound, typically as an electrical voltage. Audio signals have frequencies in the audio frequency range of roughly 20 to 20,000 Hz (the limits of human hearing) [1].

Audio signals may be synthesized directly, or may originate at a transducer such as a microphone, musical instrument pickup, phonograph cartridge, or tape head. Loudspeakers or headphones convert an electrical audio signal into sound. Digital representations of audio signals exist in a variety of formats [2, 3, 4].

References

[1] Boney, L., Tewfik, A.H., and Hamdy, K.N., "Digital Watermarks for Audio Signals," *Proceedings of the Third IEEE International Conference on Multimedia*, pp. 473-480, June 1996.

[2] Goossens, M., Mittelbach, F., Samarin, *A LaTeX Companion*, Addison-Wesley, Reading, MA, 1994.

[3] Kopka, H., Daly P.W., *A Guide to LaTeX*, Addison-Wesley, Reading, MA, 1999.

[4] Pan, D., "A Tutorial on MPEG/Audio Compression," *IEEE Multimedia*, Vol.2, pp.60-74, Summer 1998.

الطريقة رقم (2)

\cite{Boney96}

```
\begin{document}
\maketitle
\section{Introduction}
An audio signal is a representation of sound, typically as an
electrical voltage. Audio signals have frequencies in the audio
frequency range of roughly 20 to 20,000 Hz (the limits of human
hearing)~\cite{Boney96}.

Audio signals may be synthesized directly, or may originate at a
transducer such as a microphone, musical instrument pickup,
phonograph cartridge, or tape head. Loudspeakers or headphones
convert an electrical audio signal into sound. Digital
representations of audio signals exist in a variety of
formats~\cite{MG,HK,Pan}.

\begin{thebibliography}{100} % 100 is a random guess of the total
number of %references
\addtolength{\leftmargin}{0.2in} % sets up alignment with the
following line.
\setlength{\itemindent}{-0.2in}
\bibitem[Bon96]{Boney96} Boney, L., Tewfik, A.H., and Hamdy, K.N.,
``Digital
Watermarks for Audio Signals," \emph{Proceedings of the Third IEEE
International Conference on Multimedia}, pp. 473-480, June 1996.
\bibitem[Goo94]{MG} Goossens, M., Mittelbach, F., Samarin, \emph{A
LaTeX
Companion}, Addison-Wesley, Reading, MA, 1994.
\bibitem[Kop99]{HK} Kopka, H., Daly P.W., \emph{A Guide to LaTeX},
Addison-Wesley, Reading, MA, 1999.
\bibitem[Pan98]{Pan} Pan, D., ``A Tutorial on MPEG/Audio Compression,
\emph{IEEE Multimedia}, Vol.2, pp.60-74, Summer 1998.
\end{thebibliography}
\end{document}
```

1 Introduction

An audio signal is a representation of sound, typically as an electrical voltage. Audio signals have frequencies in the audio frequency range of roughly 20 to 20,000 Hz (the limits of human hearing) [Bon96].

Audio signals may be synthesized directly, or may originate at a transducer such as a microphone, musical instrument pickup, phonograph cartridge, or tape head. Loudspeakers or headphones convert an electrical audio signal into sound. Digital representations of audio signals exist in a variety of formats [Goo94, Kop99, Pan98].

References

[Bon96] Boney, L., Tewfik, A.H., and Hamdy, K.N., “Digital Watermarks for Audio Signals,” *Proceedings of the Third IEEE International Conference on Multimedia*, pp. 473-480, June 1996.

[Goo94] Goossens, M., Mittelbach, F., Samarin, *A LaTeX Companion*, Addison-Wesley, Reading, MA, 1994.

[Kop99] Kopka, H., Daly P.W., *A Guide to LaTeX*, Addison-Wesley, Reading, MA, 1999.

[Pan98] Pan, D., “A Tutorial on MPEG/Audio Compression,” *IEEE Multimedia*, Vol.2, pp.60-74, Summer 1998.

نموذج لبحث كامل تم كتابتة بأستخدام لاتك

```
\documentclass[twocolumn,showpacs,showkeys,preprintnumbers,amsma
th,amssymb]{revtex4}
\usepackage{graphicx}% Include figure files
\usepackage{dcolumn}% Align table columns on decimal point
\usepackage{bm}% bold math

\begin{document}
\title{Systematic study of even-even $^{20-32}$Mg isotopes}
\author{Fouad~A.~Majeed}\email{fouadalajeeli@yahoo.com}
\affiliation{Department of Physics, College of Education for
Pure Science, University of Babylon, P.O.Box 4., Hilla-
Babylon,Iraq}
\date{\today}
\begin{abstract}
A systematic study for 2$^+_1$ and 4$^+_1$ energies for even-
even
$^{20-32}$Mg by means of large-scale shell model calculations
using the effective interaction USDB and USDBPN with SD and SDPN
model space respectively. The reduced transition probability
{\em
B}({\em E}2;$\uparrow$) were also calculated for the chain of Mg
isotopes.Very good agreement were obtained by comparing the
first
2$^+_1$ and 4$^+_1$ levels for all isotopes with the recently
available experimental data and with the previous theoretical
work
using 3DAMP+GCM model, but studying the transition strengths
{\em
B}({\em E}2; 0$^+_{g.s.} \rightarrow$2$^+_1$) for Mg isotopes
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the experiment in detail.
\end{abstract}
\keywords{Gamma transitions and level energies, Shell model}
\pacs{23.20.Lv, 21.60.Cs}
\maketitle
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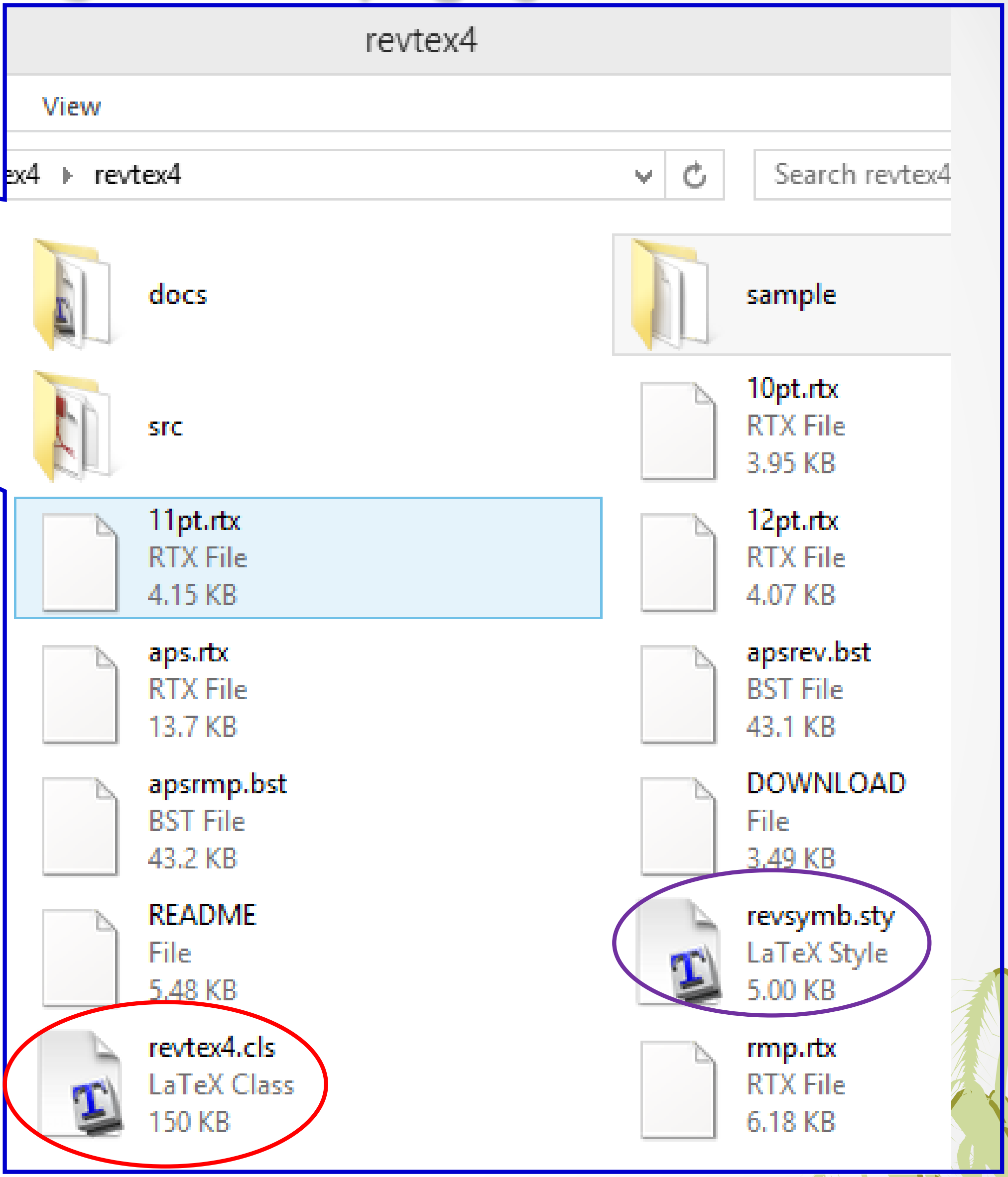

بعد تنفيذ لاتك حصلنا على التالي

Systematic study of even-even $^{20-32}$Mg isotopes

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A systematic study for 2_1^+ and 4_1^+ energies for even-even $^{20-32}$Mg by means of large-scale shell model calculations using the effective interaction USDB and USDBPN with SD and SDPN model space respectively. The reduced transition probability $B(E2;\uparrow)$ were also calculated for the chain of Mg isotopes.Very good agreement were obtained by comparing the first 2_1^+ and 4_1^+ levels for all isotopes with the recently available experimental data and with the previous theoretical work using 3DAMP+GCM model, but studying the transition strengths $B(E2;\, 0^+_{g.s.} \rightarrow 2_1^+)$ for Mg isotopes using constant proton-neutron effective charges prove the limitations of the present large-scale calculations to reproduce the experiment in detail.

PACS numbers: 23.20.Lv, 21.60.Cs
Keywords: Gamma transitions and level energies, Shell model

تابع للبحث الكامل

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\section{Introduction}
The low energy structure of magnesium nuclei has attracted
considerable interests in the last decade, both experimental and
theoretical. In particular, the sequence of isotopes $^{20-40}$Mg
encompasses three spherical magic shell numbers : N=8, 20 and 28
and, therefore presents an excellent case for studies of the
evolution of shell structure with neutron number, weakening of
spherical shell closures, disappearance of magic numbers, and the
occurrence of {\em islands of inversion} \cite{OM08}.
  Extensive experimental studies of the low-energy structure of Mg
isotopes
have been carried out at the Institute of Physical and Chemical
Research, Japan (RIKEN) \cite{H01,S09}, Michigan State University
(MSU) \cite{BV99,JM06,AL07,AG07} , the Grand Acc\'{e}l\'{e}rateur
National d'Ions Lourds, France (GANIL) \cite{VC01} and CERN
\cite{ON05,WS09}.
  In addition to numerous theoretical studies based on large-scale
shell-model calculations \cite{EC98,YU99,TO01,TT01,EC05, FM05},
the self-consistent mean-field framework, including the
nonrelativistic Hartree-Fock-Bogolibov (HFB) model with Skyrme
\cite{JT97} and Gogny forces \cite{RJ02} and the relativistic
mean-field (RMF) model \cite{SK91, ZZ96} as well as the
macroscopic-microscopic model based on a modified Nilsson
potential \cite{QJ06}, have been used to analyze the ground-state
properties (binding energies, charge radii, and deformations) and
low-lying excitation spectra of magnesium isotopes.
 The purpose of present work is to study the ground state 2$^+_1$
and
 4$^+_1$ excitation energies and the reduced transition
probabilities {\em
B}({\em E}2; 0$^+_{g.s.}\rightarrow $2$^+_1$) (e$^{2}$fm$^{4}$) of
the even-even $^{20-32}$Mg isotopes using the new version of
Nushell@MSU for windows \cite{BW07} and compare these calculations
with the most recent experimental and theoretical work.
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\section{Shell Model Calculations}
The calculations were carried out in the SD and SDPN model
spaces
with the USDB and USDBPN effective interactions \cite{BW06}
using
the shell model code  Nushell@MSU for windows \cite{BW07}.

The core was taken as $^{16}$O with 4 valence protons and
4,6,8,10,12,14,16 valence nucleons for $^{20}$Mg, $^{22}$Mg,
$^{24}$Mg, $^{26}$Mg, $^{28}$Mg, $^{30}$Mg and $^{32}$Mg
respectively distributed over 1{\em d}$_{5/2}$ , 1{\em
d}$_{5/2}$
and 2{\em s}$_{1/2}$.

The effective interaction USDB with model space SD where used in
the calculation of the $^{20-30}$Mg isotopes,  while USDBPN in
pn
formalism where employed with SDPN model space for $^{32}$Mg
nucleus.
\section{Results and Discussion}
The test of success of large-scale shell model calculations is
the
predication of the low-lying 2$^+_1$ and 4$^+_1$ and the
transition rates {\em B}({\em E}2; 0$^+_{g.s.}
\rightarrow$2$^+_1$) using the optimized effective interactions
for the description of {\em sd}-shell nuclei.

Figure 1 presents the comparison of the calculated {\em
E}$_x$(2${^+_1}$) energies from the present work (P.W.) with the
experiment \cite{NN12}, the work of J. M. Yao {\em et
al.}\cite{JM11} using 3DAMP+GCM model with the
relativistic density functional PC-F1.
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I. INTRODUCTION

The low energy structure of magnesium nuclei has attracted considerable interests in the last decade, both experimental and theoretical. In particular, the sequence of isotopes $^{20-40}$Mg encompasses three spherical magic shell numbers : N=8, 20 and 28 and, therefore presents an excellent case for studies of the evolution of shell structure with neutron number, weakening of spherical shell closures, disappearance of magic numbers, and the occurrence of *islands of inversion* [1].

Extensive experimental studies of the low-energy structure of Mg isotopes have been carried out at the Institute of Physical and Chemical Research, Japan (RIKEN) [2, 3], Michigan State University (MSU) [4–7] , the Grand Accélérateur National d'Ions Lourds, France (GANIL) [8] and CERN [9, 10].

In addition to numerous theoretical studies based on large-scale shell-model calculations [11–16], the self-consistent mean-field framework, including the non-relativistic Hartree-Fock-Bogolibov (HFB) model with Skyrme [17] and Gogny forces [18] and the relativistic mean-field (RMF) model [19, 20] as well as the macroscopic-microscopic model based on a modified Nilsson potential [21], have been used to analyze the ground-state properties (binding energies, charge radii, and deformations) and low-lying excitation spectra of magnesium isotopes.

The purpose of present work is to study the ground state 2_1^+ and 4_1^+ excitation energies and the reduced transition probabilities $B(E2; 0^+_{g.s.} \rightarrow 2_1^+)$ (e^2fm^4) of the even-even $^{20-32}$Mg isotopes using the new version of Nushell@MSU for windows [22] and compare these calculations with the most recent experimental and theoretical work.

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II. SHELL MODEL CALCULATIONS

The calculations were carried out in the SD and SDPN model spaces with the USDB and USDBPN effective interactions [23] using the shell model code Nushell@MSU for windows [22].

The core was taken as ^{16}O with 4 valence protons and 4,6,8,10,12,14,16 valence nucleons for ^{20}Mg, ^{22}Mg, ^{24}Mg, ^{26}Mg, ^{28}Mg, ^{30}Mg and ^{32}Mg respectively distributed over $1d_{5/2}$, $1d_{3/2}$ and $2s_{1/2}$.

The effective interaction USDB with model space SD where used in the calculation of the $^{20-30}$Mg isotopes, while USDBPN in pn formalism where employed with SDPN model space for ^{32}Mg nucleus.

III. RESULTS AND DISCUSSION

The test of success of large-scale shell model calculations is the predication of the low-lying 2_1^+ and 4_1^+ and the transition rates $B(E2; 0^+_{g.s.} \rightarrow 2_1^+)$ using the optimized effective interactions for the description of *sd*-shell nuclei.

Figure 1 presents the comparison of the calculated $E_x(2_1^+)$ energies from the present work (P.W.) with the experiment [24], the work of J. M. Yao *et al.*[25] using 3DAMP+GCM model with the relativistic density functional PC-F1.

The comparison shows that our calculation are in better agreement with the experiment than the work of Ref.[25]

Figure 2 shows the comparison of the calculated low-lying $E_x(4_1^+)$ excitation energies from present work (P.W.)with the experiment [24], the work of J. M. Yao *et al.*[25] using 3DAMP+GCM model with the relativistic density functional PC-F1. The comparison shows very clear that our prediction for the $E_x(4_1^+)$ are in better agreement with the experiment.

Figure 3 presents the comparison of the calculated $B(E2; 0^+_{g.s.} \rightarrow 2_1^+)$ (e^2fm^4) from present work (P.W.) with the experimental data taken from the Institute of

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